

Home Work 3
II B. Tech. (Common)
Mathematics-II (NMCI102)

Notations:

$M_{m \times n}(\mathbb{F})$: The set of all matrices of order $m \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$M_n(\mathbb{F})$: The set of all matrices of order $n \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$P(\mathbb{R})$ or $\mathbb{R}[x]$: The set of all polynomials in one variable with real coefficients.

$P_n(\mathbb{R})$ or $\mathbb{R}_n[x]$: The set of all polynomials of degree at most n in one variable with real coefficients.

$\mathbb{R}^\infty$: The set of all real-valued sequences.

$\mathcal{F}(X, \mathbb{R})$: The set of all real-valued functions on $X \subseteq \mathbb{R}$.

$C(\mathbb{R})$: The set of all real-valued continuous functions on $\mathbb{R}$

$C[0, 1]$: The set of all real-valued continuous functions defined on $[0, 1]$.

$D(0, 1)$: The set of all real-valued differentiable functions defined on $(0, 1)$.

1. Which of the following subsets are subspaces over $\mathbb{R}$ with respect to the usual addition and scalar multiplication.

- | | |
|--|---|
| (i) $\{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(x^2) = f(x)^2\}$ | (ii) $\{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(1) = 0\}$ |
| (iii) $\{A \in M_n(\mathbb{C}) \mid A \text{ is non-invertible}\}$ | (iv) $\{A \in M_3(\mathbb{R}) \mid A \text{ is a nilpotent matrix}\}$ |
| (v) $\{(x, y) \in \mathbb{R}^2 \mid (x - y)(x + y) = 0\}$ | (vi) $\{(x, y, z) \in \mathbb{R}^3 \mid x = z \}$ |
| (vii) $\{(x, y, z) \in \mathbb{R}^3 \mid x = 0 \text{ or } y = 0\}$ | (viii) $\{p(x) \in P_n(\mathbb{R}) \mid p(x) = p(1 - x)\}$ |
| (ix) $\{A \in M_n(\mathbb{C}) \mid \bar{A}^T = A\}$ | (x) $\{A \in M_n(\mathbb{C}) \mid \bar{A}^T = -A\}$ |
| (xi) $\{(x, y) \in \mathbb{R}^2 \mid y \leq x\}$ | (xii) $\{(x, y) \in \mathbb{R}^2 \mid y = x \}$ |
| (xiii) $\{(x, y) \in \mathbb{R}^2 \mid y^2 = x\}$ | (xiv) $\{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(x) = a \cos x + b \sin x + c, a, b, c \in \mathbb{R}\}$ |

Solution.

- (i) $\{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(x^2) = f(x)^2\}$

The zero function belongs to the set. Let $f(x) = x$ and $g(x) = x$. Then both satisfy the condition. However,

$$(f + g)(x) = 2x,$$

and

$$(f + g)(x^2) = 2x^2, \quad (f + g)(x)^2 = 4x^2.$$

Thus $(f + g)(x^2) \neq (f + g)(x)^2$. Hence not closed under addition.

Not a subspace

- (ii) $\{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(1) = 0\}$

The zero function satisfies $f(1) = 0$.

If $f(1) = 0$ and $g(1) = 0$, then

$$(f + g)(1) = f(1) + g(1) = 0.$$

If $c \in \mathbb{R}$,

$$(cf)(1) = cf(1) = 0.$$

Thus closed under addition and scalar multiplication.

Subspace

- (iii) $\{A \in M_n(\mathbb{C}) \mid A \text{ is non-invertible}\}$

The zero matrix is non-invertible.

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Both are non-invertible, but

$$A + B = I,$$

which is invertible.

Not a subspace

- (iv) $\{A \in M_3(\mathbb{R}) \mid A \text{ is nilpotent}\}$

The zero matrix is nilpotent.

Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Both are nilpotent, but

$$A + B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is not nilpotent.

Not a subspace

- (v) $\{(x, y) \in \mathbb{R}^2 \mid (x - y)(x + y) = 0\}$

This means $x = y$ or $x = -y$. It is the union of two lines through the origin. Not closed under addition.

Not a subspace

- (vi) $\{(x, y, z) \in \mathbb{R}^3 \mid |x| = |z|\}$

Take $(1, 0, 1)$ and $(1, 0, -1)$. Their sum is $(2, 0, 0)$, which does not satisfy the condition.

Not a subspace

- (vii) $\{(x, y, z) \in \mathbb{R}^3 \mid x = 0 \text{ or } y = 0\}$

Take $(1, 0, 0)$ and $(0, 1, 0)$. Their sum $(1, 1, 0)$ is not in the set.

Not a subspace

- (viii) $\{p(x) \in P_n(\mathbb{R}) \mid p(x) = p(1 - x)\}$

The zero polynomial satisfies the condition.

If $p(1 - x) = p(x)$ and $q(1 - x) = q(x)$, then

$$(p + q)(1 - x) = p(1 - x) + q(1 - x) = p(x) + q(x).$$

Also,

$$(cp)(1 - x) = cp(1 - x) = cp(x).$$

Subspace

- (ix) $\{A \in M_n(\mathbb{C}) \mid \overline{A}^T = A\}$

These are Hermitian matrices.

The zero matrix satisfies the condition.

If $\overline{A}^T = A$ and $\overline{B}^T = B$, then

$$\overline{(A + B)}^T = \overline{A}^T + \overline{B}^T = A + B.$$

Subspace over $\mathbb{R}$

(x) $\{A \in M_n(\mathbb{C}) \mid \overline{A}^T = -A\}$

These are skew-Hermitian matrices.

Closed under addition and real scalar multiplication.

Subspace over $\mathbb{R}$

(xi) $\{(x, y) \in \mathbb{R}^2 \mid y \leq x\}$

Take $(1, 0)$. It satisfies $0 \leq 1$.

Multiply by -1 :

$$(-1, 0),$$

which does not satisfy $0 \leq -1$.

Not a subspace

(xii) $\{(x, y) \in \mathbb{R}^2 \mid y = |x|\}$

Take $(1, 1)$.

Multiply by -1 :

$$(-1, -1),$$

but $-1 \neq |-1|$.

Not a subspace

(xiii) $\{(x, y) \in \mathbb{R}^2 \mid y^2 = x\}$

Take $(1, 1)$ and $(1, -1)$.

Their sum is $(2, 0)$, but $0^2 \neq 2$.

Not a subspace

(xiv) $\{f(x) = a \cos x + b \sin x + c \mid a, b, c \in \mathbb{R}\}$

The zero function corresponds to $a = b = c = 0$.

If

$$f(x) = a_1 \cos x + b_1 \sin x + c_1,$$

$$g(x) = a_2 \cos x + b_2 \sin x + c_2,$$

then

$$f + g = (a_1 + a_2) \cos x + (b_1 + b_2) \sin x + (c_1 + c_2).$$

Also,

$$cf = (ca_1) \cos x + (cb_1) \sin x + (cc_1).$$

Hence closed under addition and scalar multiplication.

Subspace

2. Which of the following are subspaces of the vector space $\mathbb{R}^n$?

(a) $S = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_n = 0\}$ (b) $S = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_1 + x_2 + \dots + x_n = 0\}$

(c) $S = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_1^2 + x_2^2 + \dots + x_n^2 \geq 1\}$ (d) $S = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_i = x_{n-i+1} \forall i = 1, 2, \dots, n\}$.

Solution.

- (a) $S = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_n = 0\}$

The zero vector $(0, \dots, 0)$ satisfies $x_n = 0$, so $0 \in S$.

Let $u = (x_1, \dots, x_{n-1}, 0)$ and $v = (y_1, \dots, y_{n-1}, 0)$ be in S . Then

$$u + v = (x_1 + y_1, \dots, x_{n-1} + y_{n-1}, 0),$$

so $u + v \in S$.

For $c \in \mathbb{R}$,

$$cu = (cx_1, \dots, cx_{n-1}, 0),$$

so $cu \in S$.

Hence S is a subspace of $\mathbb{R}^n$.

Subspace

- (b) $S = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_1 + \dots + x_n = 0\}$

The zero vector satisfies the condition.

Let $u = (x_1, \dots, x_n)$ and $v = (y_1, \dots, y_n)$ be in S . Then

$$x_1 + \dots + x_n = 0, \quad y_1 + \dots + y_n = 0.$$

Thus,

$$(u + v)_1 + \dots + (u + v)_n = (x_1 + y_1) + \dots + (x_n + y_n) = 0.$$

Hence $u + v \in S$.

For $c \in \mathbb{R}$,

$$c(x_1 + \dots + x_n) = 0,$$

so $cu \in S$.

Therefore S is a subspace.

Subspace

- (c) $S = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_1^2 + \dots + x_n^2 \geq 1\}$

The zero vector does not satisfy

$$0^2 + \dots + 0^2 = 0 \not\geq 1.$$

Hence $0 \notin S$.

Therefore S is not a subspace.

Not a subspace

- (d) $S = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_i = x_{n-i+1} \text{ for all } i\}$

This condition means the vector is symmetric (palindromic).

The zero vector satisfies the condition.

Let $u = (x_1, \dots, x_n)$ and $v = (y_1, \dots, y_n)$ be in S . Then

$$x_i = x_{n-i+1}, \quad y_i = y_{n-i+1}.$$

Hence,

$$(u + v)_i = x_i + y_i = x_{n-i+1} + y_{n-i+1} = (u + v)_{n-i+1}.$$

Thus $u + v \in S$.

For $c \in \mathbb{R}$,

$$(cu)_i = cx_i = cx_{n-i+1} = (cu)_{n-i+1}.$$

Hence $cu \in S$.

Therefore S is a subspace.

Subspace

3. Which of the following are subspace of $\mathbb{R}[x]$?

- (a) $S = \mathbb{R}_n[x]$ (b) $S = \{f(x) \in \mathbb{R}[x] : f(x) = f(1-x) \forall x\}$ (c) $S = \{f(x) \in \mathbb{R}[x] : f(x) = f(-x) \forall x\}$
(d) $S = \{f(x) \in \mathbb{R}[x] : f(1) \geq 0\}$ (e) $S = \{f(x) \in \mathbb{R}[x] : f'(0) + f(0) = 0\}$ (f) $S = \{f(x) \in \mathbb{R}[x] : f(x) \text{ has a root in } [-1, 1]\}$.

Solution.

- (a) $S = \mathbb{R}_n[x]$

This is the set of all polynomials of degree $\leq n$.

The zero polynomial belongs to $\mathbb{R}_n[x]$. If $f, g \in \mathbb{R}_n[x]$, then $\deg(f+g) \leq n$. If $c \in \mathbb{R}$, then $\deg(cf) \leq n$.

Hence $\mathbb{R}_n[x]$ is a subspace of $\mathbb{R}[x]$.

Subspace

- (b) $S = \{f(x) \in \mathbb{R}[x] : f(x) = f(1-x) \forall x\}$

The zero polynomial satisfies the condition.

If $f(x) = f(1-x)$ and $g(x) = g(1-x)$, then

$$(f+g)(x) = f(x) + g(x) = f(1-x) + g(1-x) = (f+g)(1-x).$$

Also, for $c \in \mathbb{R}$,

$$(cf)(x) = cf(x) = cf(1-x) = (cf)(1-x).$$

Hence S is closed under addition and scalar multiplication.

Subspace

- (c) $S = \{f(x) \in \mathbb{R}[x] : f(x) = f(-x) \forall x\}$

These are the even polynomials.

The zero polynomial is even.

If f and g are even, then

$$(f+g)(-x) = f(-x) + g(-x) = f(x) + g(x) = (f+g)(x).$$

Also,

$$(cf)(-x) = cf(-x) = cf(x).$$

Thus S is a subspace.

Subspace

- (d) $S = \{f(x) \in \mathbb{R}[x] : f(1) \geq 0\}$

The zero polynomial satisfies $f(1) = 0$.

However, $f(x) = 1$ satisfies $f(1) = 1 \geq 0$. But for scalar $c = -1$,

$$(cf)(1) = -1 < 0.$$

Hence $cf \notin S$.

Thus S is not closed under scalar multiplication.

Not a subspace

- (e) $S = \{f(x) \in \mathbb{R}[x] : f'(0) + f(0) = 0\}$

The zero polynomial satisfies $0 + 0 = 0$.

Let $f, g \in S$. Then

$$f'(0) + f(0) = 0, \quad g'(0) + g(0) = 0.$$

Hence

$$(f + g)'(0) + (f + g)(0) = f'(0) + g'(0) + f(0) + g(0) = 0.$$

Also, for $c \in \mathbb{R}$,

$$(cf)'(0) + (cf)(0) = cf'(0) + cf(0) = c(f'(0) + f(0)) = 0.$$

Thus S is a subspace.

Subspace

- (f) $S = \{f(x) \in \mathbb{R}[x] : f(x) \text{ has a root in } [-1, 1]\}$

The zero polynomial has infinitely many roots, so $0 \in S$.

Consider

$$f(x) = x - 1, \quad g(x) = -x - 1.$$

Then f has root $x = 1$, and g has root $x = -1$, both in $[-1, 1]$.

But

$$(f + g)(x) = -2,$$

which has no root in $[-1, 1]$.

Hence not closed under addition.

Not a subspace

4. Which of the following are subspaces of the vector space $M_{2 \times 2}(\mathbb{R})$?

(a) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : a + b = 0 \right\}$ (b) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : a + b + c + d = 0 \right\}$

(c) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = 0 \right\}$ (d) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : b = c = 0 \right\}$

(e) $S = \{A \in M_{2 \times 2}(\mathbb{R}) : A = A^T\}$ (f) $S = \{A \in M_{2 \times 2}(\mathbb{R}) : A = -A^T\}$

(g) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : c = 0 \right\}$ (h) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : b = 0 \right\}.$

Answers: (a) Yes (b) Yes (c) No (d) Yes (e) Yes (f) Yes (g) Yes (h) Yes.

Solution.

- (a) $a + b = 0$

The zero matrix satisfies $0 + 0 = 0$.

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix}$$

be in S . Then $a + b = 0$ and $a' + b' = 0$.

For $A + B$,

$$(a + a') + (b + b') = (a + b) + (a' + b') = 0.$$

Thus closed under addition.

For scalar k ,

$$k(a + b) = k \cdot 0 = 0.$$

Thus closed under scalar multiplication.

Subspace

(b) $a + b + c + d = 0$

Zero matrix satisfies the condition.

If $A, B \in S$, then

$$(a + a') + (b + b') + (c + c') + (d + d') = (a + b + c + d) + (a' + b' + c' + d') = 0.$$

Also,

$$k(a + b + c + d) = 0.$$

Subspace

(c) $\det A = 0$

Zero matrix has determinant 0.

However, not closed under addition.

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then $\det A = 0$ and $\det B = 0$, but

$$A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

and $\det(A + B) = 1 \neq 0$.

Not a subspace

(d) $b = c = 0$

Matrices of the form

$$\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}.$$

Zero matrix belongs to S .

Closed under addition and scalar multiplication since

$$\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} + \begin{bmatrix} a' & 0 \\ 0 & d' \end{bmatrix} = \begin{bmatrix} a + a' & 0 \\ 0 & d + d' \end{bmatrix}.$$

Subspace

(e) $A = A^T$ (symmetric matrices)

Zero matrix is symmetric.

If A and B are symmetric,

$$(A + B)^T = A^T + B^T = A + B.$$

Also,

$$(kA)^T = kA^T = kA.$$

Subspace

(f) $A = -A^T$ (skew-symmetric matrices)

Zero matrix satisfies the condition.

If A and B are skew-symmetric,

$$(A + B)^T = A^T + B^T = -A - B = -(A + B).$$

Also,

$$(kA)^T = kA^T = -kA.$$

Subspace

(g) $c = 0$

Matrices of the form

$$\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}.$$

Zero matrix belongs to S .

Closed under addition and scalar multiplication.

Subspace

(h) $b = 0$

Matrices of the form

$$\begin{bmatrix} a & 0 \\ c & d \end{bmatrix}.$$

Zero matrix belongs to S .

Closed under addition and scalar multiplication.

Subspace

5. Which of the following are subspaces of the vector space $\mathcal{F}([0, 1], \mathbb{R})$?

- (a) $S = C[0, 1]$ (b) $S = \{f \in C[0, 1] : f(0) = 0\}$ (c) $S = \{f \in C[0, 1] : f(0) = 0, f(1) = 0\}$ (d) $S = D(0, 1)$
(e) $S = \{y \in D(0, 1) : \frac{dy}{dx} + p(x)y = 0\}$ (f) $S = \{y \in D(0, 1) : \frac{dy}{dx} + p(x)y = q(x), q(x) \neq 0\}.$

Answers: (a) Yes (b) Yes (c) Yes (d) Yes (e) Yes (f) No.

Solution.

(a) $S = C[0, 1]$

The zero function is continuous on $[0, 1]$.

If $f, g \in C[0, 1]$, then $f + g$ is continuous. If $c \in \mathbb{R}$, then cf is continuous.

Thus $C[0, 1]$ is closed under addition and scalar multiplication.

Subspace

(b) $S = \{f \in C[0, 1] : f(0) = 0\}$

The zero function satisfies $0(0) = 0$.

If $f(0) = 0$ and $g(0) = 0$, then

$$(f + g)(0) = f(0) + g(0) = 0.$$

If $c \in \mathbb{R}$,

$$(cf)(0) = cf(0) = 0.$$

Thus S is closed under addition and scalar multiplication.

Subspace

- (c) $S = \{f \in C[0, 1] : f(0) = 0, f(1) = 0\}$

The zero function satisfies both conditions.

If $f, g \in S$, then

$$(f + g)(0) = 0, \quad (f + g)(1) = 0.$$

Also,

$$(cf)(0) = 0, \quad (cf)(1) = 0.$$

Hence closed under addition and scalar multiplication.

Subspace

- (d) $S = D(0, 1)$

This is the set of all differentiable functions on $(0, 1)$.

The zero function is differentiable.

If f, g are differentiable on $(0, 1)$, then $f + g$ is differentiable. If $c \in \mathbb{R}$, then cf is differentiable.

Thus $D(0, 1)$ is a subspace.

Subspace

- (e) $S = \{y \in D(0, 1) : y' + p(x)y = 0\}$

The zero function satisfies

$$0' + p(x) \cdot 0 = 0.$$

Let $y_1, y_2 \in S$. Then

$$y_1' + p(x)y_1 = 0, \quad y_2' + p(x)y_2 = 0.$$

Adding,

$$(y_1 + y_2)' + p(x)(y_1 + y_2) = y_1' + y_2' + p(x)y_1 + p(x)y_2 = 0.$$

For scalar c ,

$$(cy)' + p(x)(cy) = cy' + cp(x)y = c(y' + p(x)y) = 0.$$

Thus S is closed under addition and scalar multiplication.

Subspace

- (f) $S = \{y \in D(0, 1) : y' + p(x)y = q(x), q(x) \neq 0\}$

Suppose $y_1, y_2 \in S$. Then

$$y_1' + p(x)y_1 = q(x), \quad y_2' + p(x)y_2 = q(x).$$

Adding,

$$(y_1 + y_2)' + p(x)(y_1 + y_2) = 2q(x),$$

which is not equal to $q(x)$ (since $q(x) \neq 0$).

Hence not closed under addition.

Also, the zero function does not satisfy the equation because

$$0' + p(x) \cdot 0 = 0 \neq q(x).$$

Therefore S is not a subspace.

Not a subspace

6. For $A = \begin{bmatrix} 1 & 2 & 5 \\ 3 & 0 & 7 \\ -1 & 4 & 3 \end{bmatrix}$, examine whether $(1, 1, 1)$ and $(1, -1, 1)$ are in (a) the row space of A ; (b) the column space of A .

Answers: (a) no, yes (b) yes, no.

Solution.

Step 1: Row space of A

Row-reduce A .

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 3 & 0 & 7 \\ -1 & 4 & 3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1, \quad R_3 \rightarrow R_3 + R_1$$

$$= \begin{bmatrix} 1 & 2 & 5 \\ 0 & -6 & -8 \\ 0 & 6 & 8 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$= \begin{bmatrix} 1 & 2 & 5 \\ 0 & -6 & -8 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus a basis for the row space is

$$(1, 2, 5), \quad (0, -6, -8).$$

Hence a general element of the row space is

$$a(1, 2, 5) + b(0, -6, -8) = (a, 2a - 6b, 5a - 8b).$$

Check $v_1 = (1, 1, 1)$

We solve

$$(a, 2a - 6b, 5a - 8b) = (1, 1, 1).$$

From the first coordinate:

$$a = 1.$$

Then

$$2(1) - 6b = 1 \Rightarrow -6b = -1 \Rightarrow b = \frac{1}{6}.$$

Now check the third coordinate:

$$5(1) - 8\left(\frac{1}{6}\right) = 5 - \frac{8}{6} = 5 - \frac{4}{3} = \frac{11}{3} \neq 1.$$

Hence inconsistent.

$$\boxed{(1, 1, 1) \notin \text{Row}(A)}$$

Check $v_2 = (1, -1, 1)$

Solve

$$(a, 2a - 6b, 5a - 8b) = (1, -1, 1).$$

From first coordinate:

$$a = 1.$$

Second:

$$2(1) - 6b = -1 \Rightarrow -6b = -3 \Rightarrow b = \frac{1}{2}.$$

Third:

$$5(1) - 8\left(\frac{1}{2}\right) = 5 - 4 = 1.$$

Consistent.

$$(1, -1, 1) \in \text{Row}(A)$$

Thus for part (a):

$$(1, 1, 1): \text{ No} \quad (1, -1, 1): \text{ Yes}$$

Step 2: Column space of A

Column space of $A = \{A\mathbf{x} \mid \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3\}$. To test membership in the column space, solve

$$A\mathbf{x} = v.$$

Check $v_1 = (1, 1, 1)$

Solve

$$\begin{bmatrix} 1 & 2 & 5 \\ 3 & 0 & 7 \\ -1 & 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

This gives the system:

$$x + 2y + 5z = 1, \quad 3x + 7z = 1, \quad -x + 4y + 3z = 1.$$

Solving, one obtains a consistent solution (for example, $x = -2$, $y = -1$, $z = 1$).

Hence a solution exists.

$$(1, 1, 1) \in \text{Col}(A)$$

Check $v_2 = (1, -1, 1)$

Solve

$$A\mathbf{x} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

This yields a system which reduces to an inconsistency.

Hence no solution exists.

$$(1, -1, 1) \notin \text{Col}(A)$$

Final Answer

$$(a) \text{ Row space: } \begin{cases} (1, 1, 1) & \text{No} \\ (1, -1, 1) & \text{Yes} \end{cases}$$

$$(b) \text{ Column space: } \begin{cases} (1, 1, 1) & \text{Yes} \\ (1, -1, 1) & \text{No} \end{cases}$$

7. Determine whether $(1, 1, 1) \in L\{(1, 3, 4), (4, 0, 1), (3, 1, 2)\}$.

Solution:

Let the vectors be

$$v_1 = (1, 3, 4), \quad v_2 = (4, 0, 1), \quad v_3 = (3, 1, 2),$$

and consider whether

$$(1, 1, 1) = av_1 + bv_2 + cv_3$$

for some scalars a, b, c .

Step 1: Set up the augmented matrix

The system of equations is:

$$a + 4b + 3c = 1, \quad 3a + 0b + 1c = 1, \quad 4a + 1b + 2c = 1.$$

The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 1 \\ 3 & 0 & 1 & 1 \\ 4 & 1 & 2 & 1 \end{array} \right]$$

Step 2: Row reduce to echelon form

1. Eliminate the first column entries below the pivot:

$$R_2 \rightarrow R_2 - 3R_1, \quad R_3 \rightarrow R_3 - 4R_1$$

$$\left[\begin{array}{cccc} 1 & 4 & 3 & 1 \\ 0 & -12 & -8 & -2 \\ 0 & -15 & -10 & -3 \end{array} \right]$$

2. Make the second pivot 1 (optional) or continue elimination. Eliminate below the second pivot:

$$R_3 \rightarrow R_3 - \frac{15}{12}R_2 = R_3 - \frac{5}{4}R_2$$

Compute last row:

$$0, -15 - \frac{5}{4}(-12) = -15 + 15 = 0$$

Last augmented entry:

$$-3 - \frac{5}{4}(-2) = -3 + \frac{10}{4} = -3 + 2.5 = -0.5$$

So the REF is:

$$\left[\begin{array}{cccc} 1 & 4 & 3 & 1 \\ 0 & -12 & -8 & -2 \\ 0 & 0 & 0 & -0.5 \end{array} \right]$$

Step 3: Analyze the REF

The last row corresponds to:

$$0a + 0b + 0c = -0.5$$

which is a contradiction.

Conclusion. The system is inconsistent. Therefore,

$$(1, 1, 1) \notin L\{(1, 3, 4), (4, 0, 1), (3, 1, 2)\}.$$

8. Express the polynomial $p(t) = t^2 + 4t - 3$ over $\mathbb{R}$ as a linear combination of the polynomials $p_1(t) = t^2 - 2t + 5$, $p_2(t) = 2t^2 - 3t$ and $p_3(t) = t + 3$, if possible.

Solution:

Let us look for scalars $a, b, c \in \mathbb{R}$ such that

$$ap_1(t) + bp_2(t) + cp_3(t) = p(t).$$

Step 1: Set up the system of equations.

Expanding, we have

$$a(t^2 - 2t + 5) + b(2t^2 - 3t) + c(t + 3) = t^2 + 4t - 3$$

Combine like terms:

$$(a + 2b)t^2 + (-2a - 3b + c)t + (5a + 3c) = t^2 + 4t - 3$$

Equating coefficients gives the linear system:

$$a + 2b = 1, \quad -2a - 3b + c = 4, \quad 5a + 3c = -3.$$

Step 2: Form the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ -2 & -3 & 1 & 4 \\ 5 & 0 & 3 & -3 \end{array} \right]$$

Step 3: Row reduce to echelon form.

1. Leave R_1 as pivot row. Eliminate the first column in R_2 and R_3 :

$$R_2 \rightarrow R_2 + 2R_1, \quad R_3 \rightarrow R_3 - 5R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 6 \\ 0 & -10 & 3 & -8 \end{array} \right]$$

2. Use R_2 to eliminate the second column in R_3 :

$$R_3 \rightarrow R_3 + 10R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 13 & 52 \end{array} \right]$$

Step 4: Solve for c, b, a by back-substitution.

From the third row:

$$13c = 52 \implies c = 4$$

From the second row:

$$b + c = 6 \implies b + 4 = 6 \implies b = 2$$

From the first row:

$$a + 2b = 1 \implies a + 4 = 1 \implies a = -3$$

Step 5: Conclusion.

$$\boxed{p(t) = -3p_1(t) + 2p_2(t) + 4p_3(t)}$$

9. Write TRUE/FALSE with proper justifications.

- (a) Any set containing the zero vector is linearly dependent
- (b) If S is a linearly dependent set, then each vector in S is a linear combination of other vectors in S
- (c) Subsets of linearly independent sets are linearly independent
- (d) Subsets of linearly dependent sets are linearly dependent.

Answers: (a) TRUE (b) FALSE (c) TRUE. (d) FALSE

10. Determine the linear independence/dependence of the following sets in the corresponding vector spaces.

- (a) $\{x^3 + 2x^2, -x^2 + 3x + 1, x^3 - x^2 + 2x - 1\}$ in $P_3(\mathbb{R})$
- (b) $\{(1, 2, 2), (2, 1, 2), (2, 2, 1)\}$ in $\mathbb{R}^3$
- (c) $\left\{ \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}, \begin{bmatrix} -2 & 6 \\ 4 & -8 \end{bmatrix} \right\}$ in $M_{2 \times 2}(\mathbb{R})$
- (d) $\{(1, 2, 1, -2), (2, 5, -1, 3), (1, 3, -2, 5), (3, 1, 2, -4)\}$ in $\mathbb{R}^4$.

Solution.

(a) $\{x^3 + 2x^2, -x^2 + 3x + 1, x^3 - x^2 + 2x - 1\} \subset P_3(\mathbb{R})$

Assume a linear combination equals zero:

$$a(x^3 + 2x^2) + b(-x^2 + 3x + 1) + c(x^3 - x^2 + 2x - 1) = 0$$

Combine like terms:

$$(a + c)x^3 + (2a - b - c)x^2 + (3b + 2c)x + (b - c) = 0$$

Equating coefficients to zero gives the system:

$$a + c = 0, \quad 2a - b - c = 0, \quad 3b + 2c = 0, \quad b - c = 0.$$

$$a = 0, \quad b = 0, \quad c = 0.$$

All coefficients are zero.

Conclusion: Linearly independent.

(b) $\{(1, 2, 2), (2, 1, 2), (2, 2, 1)\} \subset \mathbb{R}^3$

Let

$$a(1, 2, 2) + b(2, 1, 2) + c(2, 2, 1) = (0, 0, 0)$$

Coordinate-wise:

$$a + 2b + 2c = 0, \quad 2a + b + 2c = 0, \quad 2a + 2b + c = 0.$$

Solve the system:

$$\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Eliminate entries below first pivot (row 1, column 1).

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 2R_1$$

Matrix becomes:

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & -3 & -2 \\ 0 & -2 & -3 \end{bmatrix}$$

Eliminate entry below second pivot (row 2, column 2)

$$R_3 \rightarrow R_3 - \frac{-2}{-3}R_2$$

REF:

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & -3 & -2 \\ 0 & 0 & -5/3 \end{bmatrix}$$

Hence $a = 0$, $b = 0$, $c = 0$.

Conclusion: Linearly independent.

(c) $\left\{ \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}, \begin{bmatrix} -2 & 6 \\ 4 & -8 \end{bmatrix} \right\} \subset M_{2 \times 2}(\mathbb{R})$ We have

$$-2 \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -2 & 6 \\ 4 & -8 \end{bmatrix}.$$

Conclusion: Linearly dependent.

(d) $\{(1, 2, 1, -2), (2, 5, -1, 3), (1, 3, -2, 5), (3, 1, 2, -4)\} \subset \mathbb{R}^4$

Let

$$a(1, 2, 1, -2) + b(2, 5, -1, 3) + c(1, 3, -2, 5) + d(3, 1, 2, -4) = (0, 0, 0, 0)$$

Coordinate-wise system:

$$a + 2b + c + 3d = 0, \quad 2a + 5b + 3c + d = 0, \quad a - b - 2c + 2d = 0, \quad -2a + 3b + 5c - 4d = 0.$$

Form augmented matrix and reduce to REF. After row reduction, we find a nontrivial solution exists (e.g., $a = 1, b = -1, c = 1, d = 0$).

Conclusion: Linearly dependent.

Answers: (a) LI (b) LI (c) LD (d) LD.

11. For what real values of k , the set $\{(k, 0, 1), (1, k + 1, 1), (1, 1, 1)\}$ is a LI subset of $\mathbb{R}^3$? [**Ans.** $k \neq 0, 1$]

Solution.

We check if

$$av_1 + bv_2 + cv_3 = 0$$

has only the trivial solution $a = b = c = 0$.

Step 1: Form the augmented matrix.

$$\begin{bmatrix} k & 1 & 1 & 0 \\ 0 & k + 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Step 2: Swap $R_1 \leftrightarrow R_3$.

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & k + 1 & 1 & 0 \\ k & 1 & 1 & 0 \end{bmatrix}$$

Step 3: Eliminate entry in row 3, column 1 using R_1 : $R_3 \rightarrow R_3 - kR_1$.

Matrix becomes:

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & k + 1 & 1 & 0 \\ 0 & 1 - k & 1 - k & 0 \end{bmatrix}$$

Step 4: Swap $R_2 \leftrightarrow R_3$.

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1-k & 1-k & 0 \\ 0 & k+1 & 1 & 0 \end{bmatrix}$$

Step 5: Eliminate entry in row 3, column 2 using pivot in row 2.

Pivot in row 2, column 2: $1-k$ (assume $k \neq 1$)

$$R_3 \rightarrow R_3 - \frac{k+1}{1-k} R_2$$

Matrix in REF:

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1-k & 1-k & 0 \\ 0 & 0 & -k & 0 \end{bmatrix}$$

Pivot in row 1: $1 \neq 0$.

Pivot in row 2: $1-k \neq 0 \implies k \neq 1$.

Pivot in row 3: $-k \neq 0 \implies k \neq 0$.

Therefore, the set is linearly independent if $k \neq 0, 1$.

12. Let u and v be distinct vectors in any vector space V over F . Show that $\{u, v\}$ is linearly dependent if and only if u or v is a multiple of the other.
13. Let $\{u, v, w\}$ be linearly independent in a real vector space V . Show that $\{\lambda u, \lambda v, \lambda w\}$, $\lambda \neq 0$, $\{u + \lambda v, v, w\}$, $\{u + v, u + w, v + w\}$, $\{u + v + w, v + w, w\}$ are also linearly independent in V and $\{u + \lambda v, v + \lambda w, w + \lambda u\}$ may not be linearly independent in V , where $\lambda \in \mathbb{R}$.
14. Let S be a subset of a vector space V over a scalar $F = \mathbb{R}$ or $\mathbb{C}$. Show that $L(S)$ is the smallest subspace of V containing S .